

From a verbal positivity check to a quantitative diagnostic

BEFORE

Observation / issue ticket

The positivity discussion was mainly verbal. It explained when an update could be negative, but did not explicitly report the simulated negative-terminal frequency.

Original conclusion

“No negative prices in a finite simulation proves only what was observed.”

Missing quantitative items:
count • denominator • frequency • uncertainty • scope

AFTER

Observed nonpositive terminal values

Method	Grids N	Observed at each grid	Frequency
EM	8–1024	0 / 160,000	0.000%
Milstein	8–1024	0 / 160,000	0.000%

With 0 events in 160,000 trials, the exact one-sided 95% binomial upper limit for the terminal frequency at any one method/grid is $1.872\text{e-}05$ (0.0019%).

Consistency with the update formulas

- EM: maximum tested one-step negative probability occurs at $N = 8$: $p = 1.189\text{e} - 21$. Zero observations are therefore unsurprising.
- Milstein: the multiplier is strictly positive for the chosen parameters, so its theoretical one-step negative probability is zero.

Scope: the saved diagnostic records terminal values only; it does not count paths that might become negative at an intermediate step and later recover.